KERALA DIPLOMA CURRICULUM REVISION 2026

Course 2331 | II | As specified in the official PDF | Source-grounded

Foundations of Robotics

Study the official course objectives in the source syllabus.

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 2331 is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 2331.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma Programme
Course code	2331
Course title	Foundations of Robotics
Semester	II
Course category	As specified in the official PDF
Periods per week	As specified in the official PDF

Item	Official information
Periods per semester	As specified in the official PDF
Credits	As specified in the official PDF
Prerequisite	Topic Course code Course Title Semester Basic knowledge in Physics 2002B Engineering Physics for Applied Electrical Technology and Computing I Basic knowledge in Chemistry 2003A Chemistry for Engineering Practices I Course Outcome CO Course Outcome Blooms Taxonomy Level CO1 Select and use sensors in various robotic application contexts Apply CO2 Select and use actuators in various robotic application contexts Apply CO3 Explain the broad concepts of robotics and its applications Understand CO4 Develop basic kind of autonomous robot Apply CO-PO mapping COURSE ARTICULATION MATRIX PROGRAM OUTCOMES
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

8 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- Study the official course objectives in the source syllabus.

Course outcomes

CO	Official outcome	Study level
CO1	3 2 - 3 2 - - - - - -	See official syllabus
CO2	2 3 3 2 2 - - - - - -	See official syllabus
CO3	2 2 - 3 3 3 - - - - 3	See official syllabus
CO4	3 3 3 2 3 3 - - - - 3 Total 10 10 6 10 10 6 - - - - 6 Correlation Level 2.5 2.5 3 2.5 2.5 3 - - - - 3 Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - "-" Teaching and Assessment Scheme Examination Scheme Teaching Scheme/Week Self-learning Total Credits Theory Practical (SS)/Week Hours/Semester C Total L T P S CIE ESE Total CIE ESE Total 4 0 0 0 6 60 4 40 60 100 0 0 0 100 Diploma Curriculum Revision 2026 2331 Page #1 L: Lecture (One unit is of one-hour duration with one credit), T: Tutorial (One unit is of one-hour duration with one credit), P: Practical (One unit is of one-hour duration with 0.5 credit), S: Project (One unit is of one-hour duration with 0.5 credit) SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination Syllabus - Major Topics Instructional Hours Module Title Major Topics L T	See official syllabus

CO–PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3 2 - 3 2 - - - - - -
CO2	CO2 2 3 3 2 2 - - - - - -
CO3	CO3 2 2 - 3 3 3 - - - - 3
CO4	CO4 3 3 3 2 3 3 - - - - 3

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Sensors in Robotics Basics of Sensors, Internal Sensors 15 0	1	15
Module 2	Actuators, Drives and Advanced Robotics Applications Building Blocks of a Robot 17 0	1	17
Module 3	Fundamentals of Robotics & Applications Definitions of Robots and Robotics 13 0	1	13
Module 4	Autonomous Robots - Line Follower and Obstacle Avoider Basic Principle of Line Follower Robot 15 0	5	As specified

MODULE 1

Sensors in Robotics Basics of Sensors, Internal Sensors 15 0

This module is presented from the official syllabus scope for Foundations of Robotics. Use the topic cards for learning and the source transcript for exact coverage.

15 **CO-linked**

official hours | see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Sensors in Robotics Basics of Sensors, Internal Sensors 15 0
- Official duration: 15
- Official topic entries captured: 1
- The full source excerpt is preserved below for coverage checking.

Sensors in Robotics Basics of Sensors, Internal Sensors 15 0

Sensors in Robotics Basics of Sensors, Internal Sensors 15 0 is an official topic in Module 1 of Foundations of Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Sensors in Robotics Basics of Sensors, Internal Sensors 15 0 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, sensors in robotics basics of sensors, internal sensors 15 0 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Sensors in Robotics Basics of Sensors, Internal Sensors 15 0 means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 1-□□ Sensors in Robotics Basics of Sensors, Internal Sensors 15 0 □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Sensors in Robotics                                                    Basics of
Sensors, Internal Sensors         15          0
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Actuators, Drives and Advanced Robotics Applications Building Blocks of a Robot 17 0

This module is presented from the official syllabus scope for Foundations of Robotics. Use the topic cards for learning and the source transcript for exact coverage.

17	**CO-linked**
official hours	see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.

- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Actuators, Drives and Advanced Robotics Applications Building Blocks of a Robot 17 0
- Official duration: 17
- Official topic entries captured: 1
- The full source excerpt is preserved below for coverage checking.

Actuators, Drives and Advanced Robotics Applications Building Blocks of a Robot 17 0

Actuators, Drives and Advanced Robotics Applications Building Blocks of a Robot 17 0 is an official topic in Module 2 of Foundations of Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Actuators, Drives and Advanced Robotics Applications Building Blocks of a Robot 17 0 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, actuators, drives and advanced robotics applications building blocks of a robot 17 0 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Actuators, Drives and Advanced Robotics Applications Building Blocks of a Robot 17 0 means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Actuators, Drives and Advanced Robotics Applications Building Blocks of a Robot 17 0 □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Actuators, Drives and Advanced Robotics Applications                           Building
Blocks of a Robot                        17           0
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Fundamentals of Robotics & Applications Definitions of Robots and Robotics 13 0

This module is presented from the official syllabus scope for Foundations of Robotics. Use the topic cards for learning and the source transcript for exact coverage.

13 official hours	**CO-linked** see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Fundamentals of Robotics & Applications Definitions of Robots and Robotics 13 0
- Official duration: 13
- Official topic entries captured: 1
- The full source excerpt is preserved below for coverage checking.

Fundamentals of Robotics & Applications Definitions of Robots and Robotics 13 0

Fundamentals of Robotics & Applications Definitions of Robots and Robotics 13 0 is an official topic in Module 3 of Foundations of Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Fundamentals of Robotics & Applications Definitions of Robots and Robotics 13 0 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, fundamentals of robotics & applications definitions of robots and robotics 13 0 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Fundamentals of Robotics & Applications Definitions of Robots and Robotics 13 0 means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 3-□□ Fundamentals of Robotics & Applications Definitions of Robots and Robotics 13 0 □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Fundamentals of Robotics & Applications
Definitions of Robots and Robotics            13          0
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Autonomous Robots - Line Follower and Obstacle Avoider Basic Principle of Line Follower Robot 15 0

This module is presented from the official syllabus scope for Foundations of Robotics. Use the topic cards for learning and the source transcript for exact coverage.

As specified	**CO-linked**
official hours	see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Autonomous Robots - Line Follower and Obstacle Avoider Basic Principle of Line Follower Robot 15 0
- Official duration: As specified
- Official topic entries captured: 5
- The full source excerpt is preserved below for coverage checking.

Autonomous Robots – Line Follower and Obstacle Avoider Basic Principle of Line Follower Robot 15 0

Autonomous Robots – Line Follower and Obstacle Avoider Basic Principle of Line Follower Robot 15 0 is an official topic in Module 4 of Foundations of Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Autonomous Robots – Line Follower and Obstacle Avoider Basic Principle of Line Follower Robot 15 0 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, autonomous robots – line follower and obstacle avoider basic principle of line follower robot 15 0 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Autonomous Robots – Line Follower and Obstacle Avoider Basic Principle of Line Follower Robot 15 0 means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Autonomous Robots – Line Follower and Obstacle Avoider Basic Principle of Line Follower Robot 15 0 □□□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

Detailed Syllabus

Detailed Syllabus is an official topic in Module 4 of Foundations of Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Detailed Syllabus using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, detailed syllabus should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Detailed Syllabus means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] Detailed Syllabus [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Mode of

Mode of is an official topic in Module 4 of Foundations of Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Mode of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, mode of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Mode of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] Mode of [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Mapped Instructional

Mapped Instructional is an official topic in Module 4 of Foundations of Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Mapped Instructional using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, mapped instructional should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Mapped Instructional means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] Mapped Instructional [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Subtopic Student Learning Outcome delivery Taxonomy Level

Subtopic Student Learning Outcome delivery Taxonomy Level is an official topic in Module 4 of Foundations of Robotics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Subtopic Student Learning Outcome delivery Taxonomy Level using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, subtopic student learning outcome delivery taxonomy level should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Subtopic Student Learning Outcome delivery Taxonomy Level means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 4-□□ Subtopic Student Learning Outcome delivery Taxonomy Level □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Autonomous Robots – Line Follower and Obstacle Avoider                     Basic
Principle of Line Follower Robot       15           0

Detailed Syllabus

Mode of

Mapped                             Instructional
             Subtopic                              Student Learning Outcome
delivery                  Taxonomy Level

CO                                   Hours

(L/T)
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

- for Self-Learning
- Week Self-Learning description Hours Module I
- Prepare notes on internal sensors such as position, velocity sensors, and accelerometers with applications. 6
- Study external sensors including proximity, force, and torque sensors and identify their real-time uses in robotics. 6
- Create a table mapping different sensing variables (temperature, light, humidity, sound) with suitable sensors. 6
- Prepare chart for all sensors and take a seminar each sensors 6 Module II
- Study different types of electric motors (DC, servo, stepper) used as actuators in robotic systems. 6
- Apply speed and direction control techniques of motors and selection criteria for actuators in robots 6
- Case study on selection of sensors and actuators for self-driving cars and maze-solving robots (advanced content). 6 Module III
- Study and summarize the definitions, need, types, and characteristics of robots using textbook and NPTEL resources. 6
- Prepare a comparative chart on advantages and limitations of robots in different application areas such as medical, defense,
- and domestic sectors.
- Explore real-world industrial robot applications (material handling, welding, painting) through videos and case studies. 6
- Prepare report on case studies based robot applications 6 Module IV
- Study the basic principle of line follower robots and color sensing using online simulations and reference videos. 6
- Identify different sensors used in line follower and obstacle avoiding robots and explain their working principles. 6
- Analyze the block diagram of a line follower robot and explain signal flow from sensor to actuator 6
- Total Self-Learning hours 90 Learning Resources TEXT BOOK
- Title Author Publication
- Robotics for Engineers Yoram Koren McGraw-Hill Education

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
— Theory
                                                 CA1                  CA2
CA3                CA4                    CA5            CA6           ESE


     Mode          Attendance                   Self-learning    Self-learning
Self-learning    Self-learning                   Series      Series           Written
                                                  activities        activities

activities       activities                     Exam        Exam                Exam
                                                  (Module 1)        (Module 2)
(Module 3)       (Module 4)                   (2 modules) (2 modules)         (theory)
   Duration
2 hours         2 hours       2.5 hours
  Exam mark                                               15                  15
15                   15                  50            50            60
 Converted to
20              20
     Marks                 5                                                       15
20                    60
     Total
40                                                                         60
   Tentative           End of
End of
                                            By 4th week       By 7th week           By
11th week      By 14th week               8th week        15th week
   schedule           semester
semester
```

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain Sensors in Robotics Basics of Sensors, Internal Sensors 15 0. (3 marks)

Answer: Begin with the standard definition or identity of *Sensors in Robotics Basics of Sensors, Internal Sensors 15 0*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 2

Define or explain Actuators, Drives and Advanced Robotics Applications Building Blocks of a Robot 17 0. (3 marks)

Answer: Begin with the standard definition or identity of *Actuators, Drives and Advanced Robotics Applications Building Blocks of a Robot 17 0*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 3

Define or explain Fundamentals of Robotics & Applications Definitions of Robots and Robotics 13 0. (3 marks)

Answer: Begin with the standard definition or identity of *Fundamentals of Robotics & Applications Definitions of Robots and Robotics 13 0*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 4

Define or explain Autonomous Robots – Line Follower and Obstacle Avoider Basic Principle of Line Follower Robot 15 0. (3 marks)

Answer: Begin with the standard definition or identity of *Autonomous Robots – Line Follower and Obstacle Avoider Basic Principle of Line Follower Robot 15 0*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Detailed Syllabus. (3 marks)

Answer: Begin with the standard definition or identity of *Detailed Syllabus*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Mode of. (3 marks)

Answer: Begin with the standard definition or identity of *Mode of*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Mapped Instructional. (3 marks)

Answer: Begin with the standard definition or identity of *Mapped Instructional*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **Sensors in Robotics Basics of Sensors, Internal Sensors 15 0**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Actuators, Drives and Advanced Robotics Applications Building Blocks of a Robot 17 0**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Fundamentals of Robotics & Applications Definitions of Robots and Robotics 13 0**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **Autonomous Robots – Line Follower and Obstacle Avoider Basic Principle of Line Follower Robot 15 0**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- URL Modules Covered
- https://nptel.ac.in NPTEL Robotics Courses
- https://create.arduino.cc/projecthub Arduino Project

IN-PAGE SEARCH

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